

# Rapid Solution of Problems by Quantum Computation

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## 1. ABSTRACT & THEORETICAL FOUNDATION

We prove that quantum computers can evaluate global properties of functions, determining whether a Boolean function is constant or balanced in a single quantum evaluation compared to exponential classical queries.

## 2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"If you have a coin that might be normal (half heads, half tails) or double-headed, a classical computer might need to flip it over a million times to be 100% sure. A quantum computer knows with just ONE flip."

## 3. Mathematical Formulation & Unitary Rule

$$|\psi_{\text{final}}\rangle = \sum_{z \in \{0,1\}^n} \left[ \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} \right] |z\rangle$$

For  $z = 0^n$ , amplitude is  $1/2^n * \sum (-1)^{f(x)}$ . For constant  $f$ , amplitude is  $\pm 1$ ; for balanced  $f$ , sum evaluates to exactly 0.

## 4. Step-by-Step Circuit Sequence

$H^{\otimes n}$  on inputs,  $X+H$  on ancilla, Oracle  $U_f$ ,  $H^{\otimes n}$  on inputs, measure all  $n$  inputs.

## 5. Executable IBM Qiskit (Python) Code

```
from qiskit import QuantumCircuit

n = 3
qc = QuantumCircuit(n + 1, n)

# Prepare input in  $|+\rangle^n$  and ancilla in  $|-\rangle$ 
qc.x(n)
qc.h(range(n + 1))
```

## 6. Computational Complexity & Speedup

| Classical Time:                          | Quantum Time:                                     | Speedup Class: |
|------------------------------------------|---------------------------------------------------|----------------|
| $O(2^{n-1} + 1)$ queries (deterministic) | $O(1)$ (single oracle query, exact deterministic) | Exponential    |

## 7. Key Applications & Future Research Directions

- Separation of complexity classes EQP and P
- Demonstrating global interference across  $2^n$  branches
- Quantum query complexity theory

### Future Work:

Extension to generalized promise problems and hidden subgroup problems over non-abelian groups.